Broad-Spectrum Antibiotic Prescribing and Regional Antimicrobial Resistance

Temporal Lag, Regional Heterogeneity, and Interpretable Prediction

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Abstract

Antimicrobial resistance (AMR) can respond slowly and unevenly to changes in antibiotic use. This project studies three related questions: whether lower broad-spectrum prescribing is followed by lower resistance, how long the response may take, and why regional trajectories differ. The analysis combines an area-quarter panel, temporal lag models, regional trajectory clustering, a time-respecting random-forest model, SHAP interpretation, and a peer-based prescribing scenario. The simulation analysis contains 40 regional series from April 2021 to April 2026 and produces 840 area-quarter observations. Four distinct regional response patterns are recovered, the clearest positive lag association appears at three quarters, and the final temporal holdout reaches balanced accuracy 0.722 and ROC-AUC 0.710. These results show how statistical analysis and interpretable machine learning can be combined to study delayed and heterogeneous AMR responses.

1 Background and Motivation

My interest in antibiotic safety began with a family experience. My mother lost her hearing in childhood after receiving gentamicin, and my family has long understood the episode as a drug-related adverse effect. Aminoglycosides such as gentamicin are known to cause vestibular or cochlear ototoxicity, which can sometimes be permanent [1]. Growing up with this history made me curious about a broader question: how can antibiotics be used effectively while limiting the harms that may accompany their use? This question gradually led me from individual drug safety to antibiotic stewardship and antimicrobial resistance (AMR) at the population level.

Antibiotics remain essential for treating bacterial infections, but their use also creates selective pressure that can favor resistant organisms. Broad-spectrum antibiotics are particularly important in stewardship because they act against a wide range of bacteria and are generally reserved for situations in which narrower-spectrum treatments may not be appropriate [4]. At the population level, the relationship between prescribing and resistance may unfold over time rather than appear immediately. Aliabadi et al. linked primary-care prescribing with *Escherichia coli* bacteraemia resistance in England and, based on previous ecological evidence, considered effects of prescribing changes within a six-month period [2]. This temporal structure motivates looking beyond same-period correlations and asking when changes in resistance become visible after prescribing changes.

Regional variation adds another layer to the problem. Observational studies have associated resistant *E. coli* bacteraemia with factors including age, comorbidity, socioeconomic deprivation,

and previous antimicrobial exposure [3]. This suggests that similar reductions in broad-spectrum prescribing may be followed by different resistance trajectories across areas. Public data sources make this question suitable for regional analysis: OpenPrescribing reports broad-spectrum prescribing measures, the NHS Business Services Authority publishes detailed community prescribing records, and the UK Health Security Agency reports resistance and testing measures for *E. coli* bacteraemia [4, 6, 5].

Against this background, I focus on three questions. What kinds of prescribing–resistance trajectories appear across regions? At what time lag is prescribing most strongly associated with later resistance? Can subsequent resistance improvement be predicted from recent prescribing history together with demographic, socioeconomic, disease-burden, and testing information?

2 Methods

2.1 Simulation analysis and panel construction

The simulation analysis follows the same variable structure as the intended public-data study. Monthly prescribing records, AMR surveillance measures, resistance-testing volume, and annual regional covariates are generated for 40 regions. The study period runs from April 2021 to April 2026. Broad-spectrum prescribing is measured as prescription items per 1,000 registered patients, and the primary resistance outcome is the percentage of tested *E. coli* bacteraemia isolates resistant to co-amoxiclav.

The monthly series are converted into non-overlapping quarterly observations. Resistance values are retained only when testing volume meets the minimum testing threshold. Annual covariates are then joined by region and year. These variables include the share of residents aged 65 and 80 years or older, deprivation score, and prevalence of diabetes, chronic kidney disease, chronic obstructive pulmonary disease, and cancer. The final panel contains 840 observations from 40 regions over 21 quarters. Fourteen data-quality checks covering required fields, date structure, duplicate keys, numerical ranges, and covariate coverage all pass before analysis.

2.2 Regional trajectories and temporal lag

Regional heterogeneity is summarized in two ways. A transparent change analysis compares the first and last four-quarter windows and assigns each region to one of four patterns: prescribing and resistance both decline; prescribing declines while resistance does not; resistance declines while prescribing does not; or neither improves. Dynamic time warping (DTW) clustering is then applied to the joint prescribing-resistance trajectories without using these labels.

Temporal association is examined with a distributed-lag fixed-effects model:

$$R_{it} = \sum_{k=1}^{4} \beta_k P_{i,t-k} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where R_{it} is resistance in region i and quarter t, $P_{i,t-k}$ is standardized prescribing lagged by k quarters, α_i controls for persistent regional differences, and δ_t captures common quarter effects. Observations are weighted by resistance-testing volume. An interrupted time-series analysis provides an additional summary of the prescribing trend around the study intervention point.

2.3 Prediction, interpretation, and scenario analysis

The prediction task asks whether resistance falls by at least 0.5 percentage points four quarters later. A random forest uses current prescribing, recent prescribing changes and slopes, current resistance, testing measures, age structure, deprivation, and chronic-disease prevalence. Evaluation follows chronological order. Earlier quarters are used for training and later quarters for testing through rolling-origin evaluation, followed by a final four-quarter holdout.

Balanced accuracy and ROC-AUC summarize discrimination, while average precision, F1, precision, recall, and Brier score provide complementary information. Feature importance and SHAP values are used to inspect the fitted model. The final scenario analysis groups similar regions by demographic, socioeconomic, disease-burden, and resistance profiles. For regions above their peer-group median prescribing level, the prescribing input is lowered to that median and the predicted probability is recalculated.

3 Results

3.1 Overall trend and regional response patterns

The panel contains 840 area-quarter observations. Figure 1 shows that mean prescribing and mean resistance follow different temporal shapes. Their movements are not synchronized quarter by quarter, which motivates explicit lag analysis.

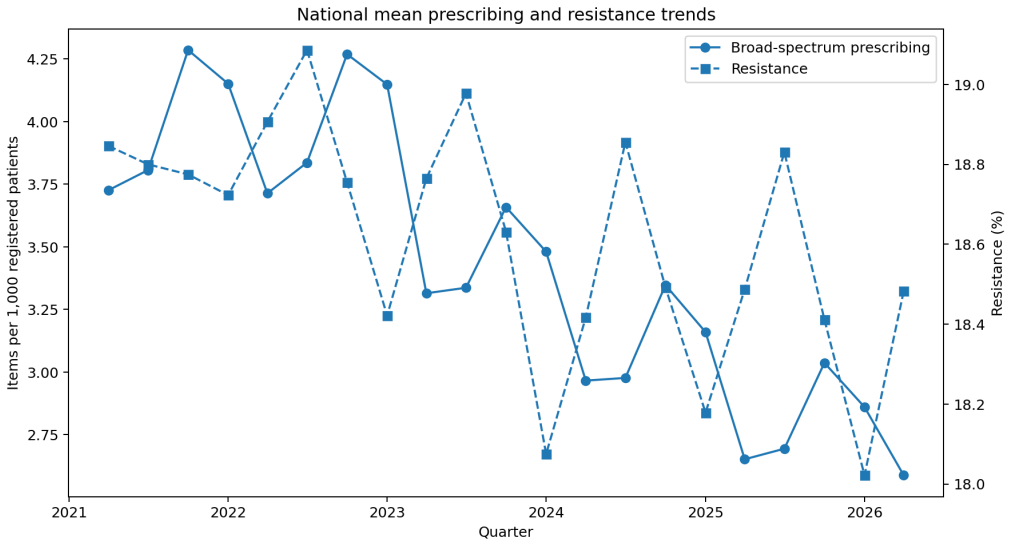
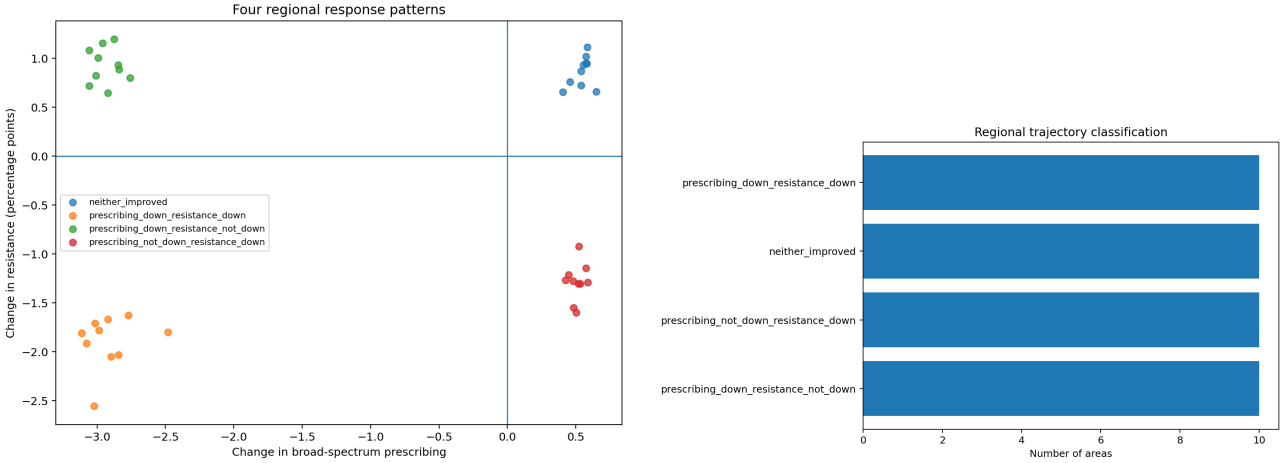


Figure 1: Mean broad-spectrum prescribing and resistance across the 40 regions.

The four regional response classes are evenly represented, with 10 regions in each group. Figure 2 shows that areas with similar changes in prescribing can still experience different resistance changes. DTW clustering produces a silhouette score of 0.490 and an adjusted Rand index of 1.000 against the known trajectory groups, indicating that the joint time-series patterns are recovered clearly.



(a) Change in prescribing and resistance. **(b)** Number of regions in each response class.

Figure 2: Regional heterogeneity in prescribing-resistance trajectories.

3.2 Temporal lag analysis

The interrupted time-series analysis estimates an immediate prescribing level change of −0.403 items per 1,000 patients at the study intervention point, followed by a monthly slope change of −0.025. The distributed-lag estimates in Figure 3 show a more detailed temporal pattern. The one- and two-quarter coefficients are small and statistically uncertain. The three-quarter coefficient is positive at 0.686 (95% CI 0.363 to 1.008), while the four-quarter coefficient is negative at −0.471 (95% CI −0.857 to −0.085).

This pattern shows that the relationship between prescribing and later resistance depends on the observation window. A single same-quarter correlation would miss this temporal structure.

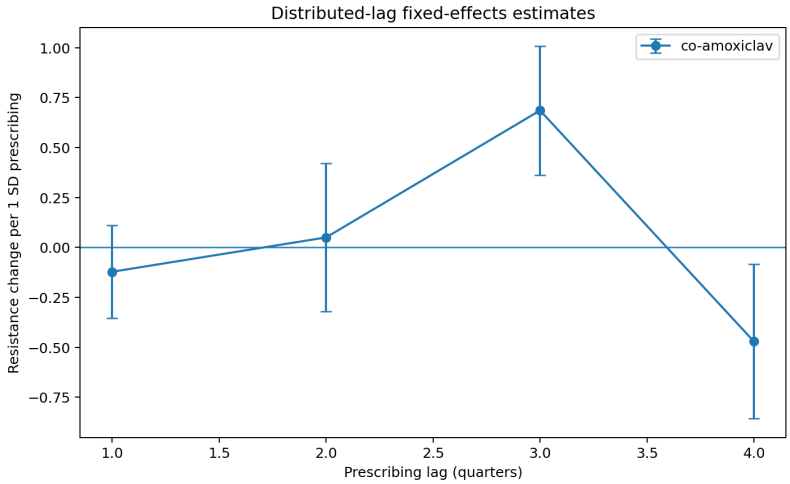


Figure 3: Distributed-lag fixed-effects estimates with 95% confidence intervals.

3.3 Predictive performance and model interpretation

Performance varies across the four rolling-origin evaluation windows. ROC-AUC ranges from 0.548 to 0.947 and balanced accuracy ranges from 0.472 to 0.875. The final temporal holdout

contains 87 eligible observations, including 15 resistance-improvement outcomes. At the selected probability threshold of 0.20, the model achieves balanced accuracy 0.722 and ROC-AUC 0.710. Recall is 1.000, precision is 0.273, F1 is 0.429, average precision is 0.261, and the Brier score is 0.147.

Table 1: Final temporal holdout performance.

Metric	Value	Metric	Value
Balanced accuracy	0.722	ROC-AUC	0.710
F1	0.429	Average precision	0.261
Precision	0.273	Recall	1.000
Brier score	0.147	Threshold	0.200

Figure 4 compares predicted probabilities with observed improvement rates. The model separates higher- and lower-risk observations, while the deviations from the diagonal indicate that probability estimates are not perfectly calibrated.

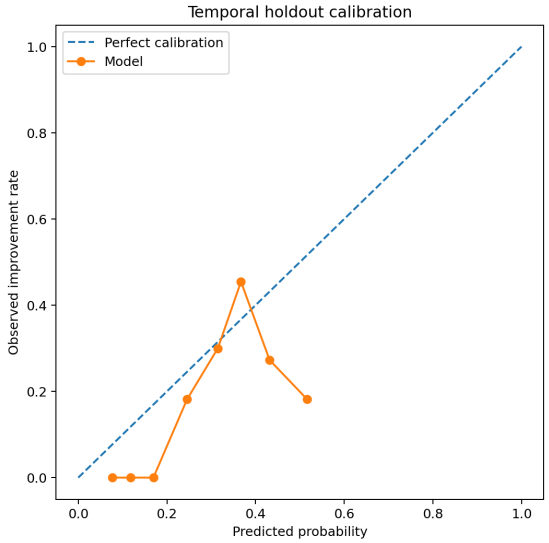
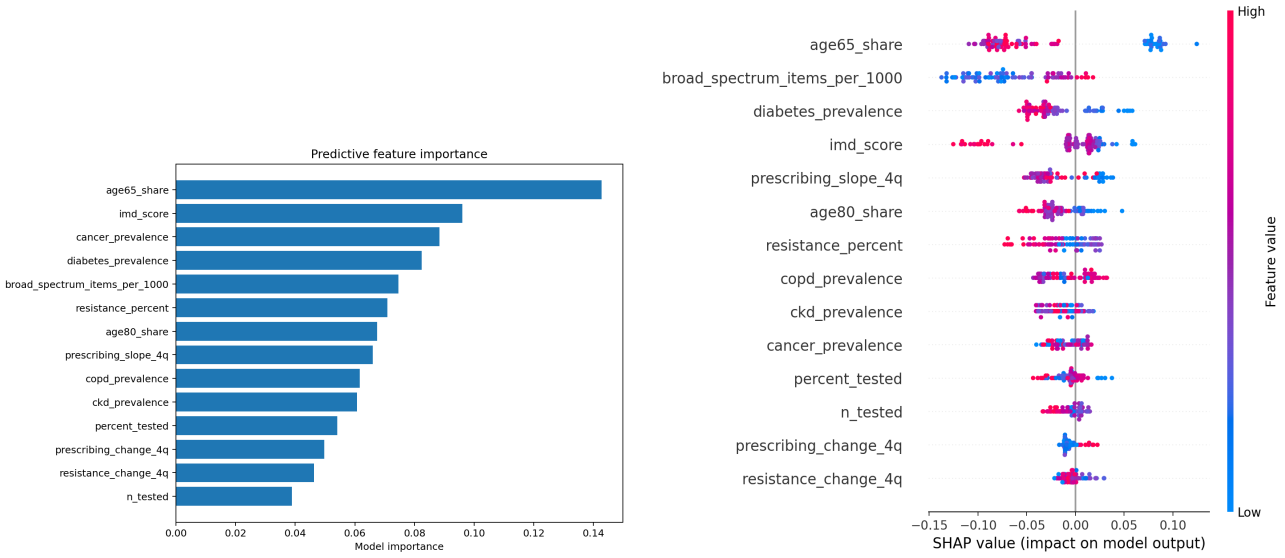


Figure 4: Calibration on the final temporal holdout.

The random-forest ranking places age-65 share first, followed by deprivation score, cancer prevalence, and diabetes prevalence. SHAP analysis also ranks age-65 share highly and places current broad-spectrum prescribing among the most influential variables. The two views suggest that both current prescribing and regional context contribute to the prediction function.



(a) Random-forest feature importance. **(b)** SHAP summary.

Figure 5: Interpretation of the final prediction model.

3.4 Peer-median prescribing scenario

The peer-based scenario lowers prescribing in 19 of the 40 regions. The predicted probability changes vary across areas, ranging approximately from −0.298 to +0.031. Figure 6 shows the regions with the largest absolute changes. The spread in these results reflects the fact that the fitted model combines prescribing with the broader regional profile rather than applying one common response to every area.

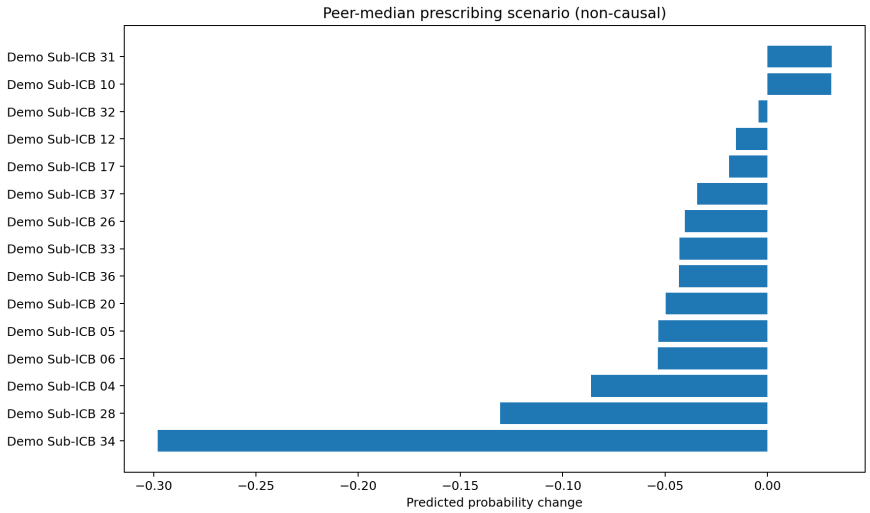


Figure 6: Predicted probability change under the peer-median prescribing scenario.

4 Conclusion

The project develops a framework for studying delayed and heterogeneous relationships between broad-spectrum antibiotic prescribing and AMR. The regional trajectory analysis shows four qualitatively different response patterns, and DTW clustering recovers the same underlying structure from the time series. The lag analysis shows that prescribing-resistance associations change across one- to four-quarter windows, with the clearest positive estimate at three quarters.

The predictive analysis provides a complementary view. A time-respecting random forest achieves moderate discrimination on the final holdout, while SHAP and feature importance indicate that both prescribing history and regional characteristics are informative. The peer-median scenario then translates the fitted model into a regional comparison that highlights heterogeneous predicted responses. Together, these components show where, when, and under what local conditions resistance improves after prescribing changes.

References

[1] J. A. Leis, J. A. Rutka, and W. L. Gold. Aminoglycoside-induced ototoxicity. *CMAJ*, 187(1):E52, 2015. DOI: 10.1503/cmaj.140339.

[2] S. Aliabadi, P. Anyanwu, E. Beech, et al. Effect of antibiotic stewardship interventions in primary care on antimicrobial resistance of *Escherichia coli* bacteraemia in England (2013–18): a quasi-experimental, ecological, data linkage study. *The Lancet Infectious Diseases*, 21(12):1689–1700, 2021. DOI: 10.1016/S1473-3099(21)00069-4.

[3] C. McCowan, A. Bakhshi, A. McConnachie, et al. *E. coli* bacteraemia and antimicrobial resistance following antimicrobial prescribing for urinary tract infection in the community. *BMC Infectious Diseases*, 22:805, 2022. DOI: 10.1186/s12879-022-07768-7.

[4] OpenPrescribing. Antibiotic stewardship: co-amoxiclav, cephalosporins & quinolones prescribing volume. https://openprescribing.net/measure/broad_spectrum_list/definition/. Accessed September 30, 2026.

[5] UK Health Security Agency. Antimicrobial resistance in *Escherichia coli* (*E. coli*) bacteraemia. https://ukhsa-dashboard.data.gov.uk/antimicrobial-resistance/antimicrobial-resistance-in-ecoli-bacteraemia. Accessed September 30, 2026.

[6] NHS Business Services Authority. English Prescribing Dataset (EPD) with SNOMED Code. NHSBSA Open Data Portal. https://opendata.nhsbsa.net/dataset/english-prescribing-dataset-epd-with-snomed-code. Accessed September 30, 2026.